Multi-Agent Reinforcement Learning for Joint Node Selection and Power Allocation in Cognitive Tracking Radar Networks

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***Abstract*—This paper investigates joint node selection and power allocation for multistatic cognitive tracking radar networks. A unified framework integrating target tracking and posterior Cramér-Rao lower bound (PCRLB)-based performance modeling analysis is developed to enable distributed sensing decisions without explicit inter-node communication. The problem is formulated as a stochastic Markov game (SMG) and addressed using a centralized training and decentralized execution (CTDE) reinforcement learning approach based on multi-agent proximal policy optimization (MAPPO). Simulation results demonstrate that the proposed framework achieves effective resource management in multistatic radar networks while enabling low-overhead distributed execution during deployment.**

***Index Terms*—Target tracking, cognitive radar, resource management, game theory, multi-agent reinforcement learning**

I. Introduction

Target tracking is a core function in radar sensing systems [1]. Compared with monostatic radar systems, multistatic radar networks provide enhanced coverage, improved imaging resolution [2], and greater interference resilience via spatial diversity and node redundancy [3], [4]. However, tracking performance critically depends on effective task assignment among radar nodes [5]–[7], while practical constraints further motivate efficient resource allocation strategies [8], [9].

Cognitive radar systems adapt sensing strategies based on environmental feedback to enhance situational awareness and efficiency [10], [11]. In multistatic cognitive radar networks, joint optimization of node selection and power allocation is crucial for system-wide tracking performance. Unlike single-radar optimization, this requires coordinated decisions across distributed nodes, increasing resource coupling, and thus is naturally formulated as a sequential decision-making problem rather than a static optimization task.

Markov Decision Processes (MDPs) provide a classical framework for single player sequential decision-making under uncertainty [12], [13]. When multiple players interact, the problem is more accurately modeled as an SMG [14], [15]. To solve the resulting SMG, reinforcement learning (RL) techniques can be employed, with Nash equilibrium (NE) representing stable solutions in which no agent can unilaterally improve its payoff [16]. RL enables agents to learn optimal policies through interaction with the environment [17]–[19].

In multi-player settings, multi-agent reinforcement learning (MARL) extends RL to environments in which state transitions and rewards depend on joint actions, making it particularly suitable for distributed radar systems. Among MARL methods, MAPPO-based approaches have been shown to achieve competitive or superior performance on several representative cooperative MARL benchmarks with minimal hyperparameter tuning [20]. Moreover, MAPPO-based architectures have also been successfully extended to hybrid discrete-continuous action spaces, demonstrating effective training and improved performance in practical multi-agent decision problems [20], [21].

Related work has explored game-theoretic and learning-based approaches to radar resource management, including MDP-based node selection [22], non-cooperative power control games [23], game-theoretic power allocation in multistatic radar networks [24], and joint detection-threshold/illumination-time optimization under clutter [25]. These studies have demonstrated the value of sequential optimization and learning for radar resource management. However, most existing methods either optimize only part of the decision process, rely on centralized global information during deployment, or do not explicitly couple decentralized decision making with PCRLB-based tracking-performance modeling.

Motivated by these limitations, this paper develops a unified framework for joint node selection and power allocation in multistatic cognitive tracking radar networks. The main idea is to model the distributed sensing problem as an SMG and to solve it through a centralized-training and decentralized-execution MAPPO scheme. In summary, this paper contributes a unified PCRLB-driven SMG formulation and a CTDE-MAPPO solution for joint node selection and power allocation in multistatic cognitive tracking radar networks.

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II. System Model

Consider a multistatic radar network with N radar nodes, whose positions are denoted as (x_i, y_i), where $i = 1, ..., N$. The scenario is shown in Fig.1.

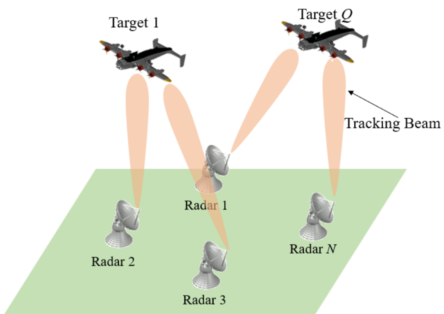

Fig. 1. Multistatic Tracking Radar Network.

This radar network with N radar station nodes is designed to track Q point targets in a cluttered environment. To keep the focus on resource allocation rather than low-level detection and data association, we assume that each radar node performs local preprocessing within each scan and outputs, for each target, at most one validated range–bearing measurement.

A. Target State Model

The target motion is modeled using a nearly constant acceleration (NCA) model [26]. The dynamic target model of the q-th target at the k-th time step can be defined as

$$\mathbf{X}_k^q = \mathbf{F}\mathbf{X}_{k-1}^q + \mathbf{V}_k^q, \tag{1}$$

where $\mathbf{X}_k^q$ represents the state of q-th target at k-th time step. It is defined as

$$\mathbf{X}_k^q = [x_k^q, \dot{x}_k^q, \ddot{x}_k^q, y_k^q, \dot{y}_k^q, \ddot{y}_k^q]^{\mathrm{T}}, \tag{2}$$

$\mathbf{F}$ represents the state transition matrix, defined as

$$\mathbf{F} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & \Delta T & \Delta T^2/2 \\ 0 & 1 & \Delta T \\ 0 & 0 & 1 \end{bmatrix}, \tag{3}$$

where $\otimes$ represents the Kronecker product. $\mathbf{V}_k^q$ represents the zero-mean Gaussian process noise of the q-th target at the k-th time step, and its covariance matrix is defined as

$$\mathbf{Q}_k^q = \kappa \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} \Delta T^5/20 & \Delta T^4/8 & \Delta T^3/6 \\ \Delta T^4/8 & \Delta T^3/3 & \Delta T^2/2 \\ \Delta T^3/6 & \Delta T^2/2 & \Delta T \end{bmatrix}, \tag{4}$$

where κ represents the process noise intensity.

B. Measurement Model

The measurement $\mathbf{Z}_k^q$ can be defined as

$$\mathbf{Z}_k^q = h(\mathbf{X}_k^q) + \mathbf{W}_k^q, \tag{5}$$

where $h(\mathbf{X}_k^q)$ represents the measurement function, which can be defined as

$$h(\mathbf{X}_k^q) = \begin{bmatrix} r_k^q \\ \theta_k^q \end{bmatrix} = \begin{cases} \sqrt{(x_k^q - x_i)^2 + (y_k^q - y_i)^2} \\ \arctan(\frac{y_k^q - y_i}{x_k^q - x_i}) \end{cases}, \tag{6}$$

where $r_{i,k}^q \geq 0$ denotes the measured range, and $\theta_{i,k}^q \in (-\pi, \pi]$ denotes the bearing angle.

The measurement noise $\mathbf{W}_k^q$ is assumed to be zero mean Gaussian distributed, with covariance matrix $\mathbf{R}_k^q$, defined as

$$\mathbf{R}_k^q = \begin{bmatrix} \sigma_{r_k^q}^2 & 0 \\ 0 & \sigma_{\theta_k^q}^2 \end{bmatrix}, \tag{7}$$

For the MARL policy input, the local observation of radar node i at time step k is represented by a target-conditioned fixed-length vector

$$\mathbf{o}_{i,k} = \left[(\tilde{\mathbf{z}}_{i,k}^1)^T, b_{i,k}^1, \ldots, (\tilde{\mathbf{z}}_{i,k}^Q)^T, b_{i,k}^Q\right]^T, \tag{8}$$

where $\tilde{\mathbf{z}}_{i,k}^q \in \mathbb{R}^2$ denotes the validated range–bearing measurement associated with target q at radar node i, and $b_{i,k}^q \in \{0, 1\}$ is an availability indicator. If target q is not validated at node i during scan k, then $b_{i,k}^q = 0$ and $\tilde{\mathbf{z}}_{i,k}^q$ is set to a null placeholder. This abstraction isolates the resource-allocation layer from the low-level measurement-origin uncertainty, while preserving a decentralized observation structure.

C. Posterior Cramér-Rao Lower Bound

In target tracking scenarios, the PCRLB provides a tight bound for the mean-square error of any unbiased estimator and is independent of the specific filtering algorithm. Moreover, this bound is amenable to algebraic manipulation and predictive analysis [27].

We define $\mathbf{C}_k^q$ as the error covariance matrix of the q-th target at the k-th step. The PCRLB is defined to be the inverse of the Bayesian Fisher information matrix (BFIM) of the q-th target at the k-th step $\mathbf{J}(\mathbf{X}_k^q)$. The following relation holds

$$\mathbf{C}_k^q \triangleq \mathbb{E}\left[\left(\mathbf{X}_k^q - \hat{\mathbf{X}}_k^q\right)\left(\mathbf{X}_k^q - \hat{\mathbf{X}}_k^q\right)^T\right] \geq \mathbf{J}_{k,q}^{-1}, \tag{9}$$

where $\hat{\mathbf{X}}_k^q$ is the estimated state.

The BFIM is composed of two terms [28]

$$\mathbf{J}(\mathbf{X}_k^q) = \mathbf{J}^P(\mathbf{X}_k^q) + \mathbf{J}^D(\mathbf{X}_k^q), \tag{10}$$

where $\mathbf{J}^P(\mathbf{X}_k^q)$ is the Fisher information matrix (FIM) of prior information, and $\mathbf{J}^D(\mathbf{X}_k^q)$ is the FIM of data, which can be calculated as

$$\begin{cases} \mathbf{J}^P(\mathbf{X}_k^q) = \left[\mathbf{Q}_k^q + \mathbf{F} \cdot \mathbf{J}^{-1}(\mathbf{X}_{k-1}^q) \cdot \mathbf{F}^{\mathbf{T}}\right]^{-1} \\ \mathbf{J}^D(\mathbf{X}_k^q) = \sum_{i=1}^N \mathbb{E}_{\mathbf{X}_k^q}\left[\delta_{i,k}^q \zeta_{i,k}^q (\mathbf{H}_{i,k}^q)^T (\mathbf{R}_{i,k}^q)^{-1} \mathbf{H}_{i,k}^q\right] \end{cases}, \tag{11}$$

where $\mathbf{H}_{i,k}^q$ is the Jacobian matrix of the measurement function $h(\mathbf{X}_k^q)$, $\delta_{i,k}^q \in \{0, 1\}$ indicates whether radar node i is assigned to target q, and $\zeta_{i,k}^q \in [0, 1]$ is the information reduction factor (IRF) [28].

III. Stochastic Markov Game and Multi-Agent Reinforcement Learning Algorithms

A. Stochastic Markov Game

A Markov Decision Process (MDP) provides a standard framework for single player sequential decision-making [29].

An SMG could be regarded as a multi-player extension version of MDP. The central idea of game theory is to model a strategic resource allocation problem as a game between a set of players [30], [31]. An SMG is defined by the tuple $\langle N, S, \{\mathcal{A}_i\}_{i=1}^N, P, \{R_i\}_{i=1}^N, \gamma\rangle$, where N denotes the number of players, S is the set of environmental states shared by all players, $\mathcal{A}_i$ is the action space of player i, P is the state transition probability, R_i is the reward function of player i, and $\gamma \in [0,1]$ is the discount factor.

In the considered multistatic radar tracking scenario, the number of players equals the number of radar nodes (N), and the state at time step k is defined as the joint target state $s_k \triangleq \{\mathbf{X}_k^q\}_{q=1}^Q$. The action taken by player i at time step k is denoted by

$$a_{i,k} \triangleq \big(\mathbf{p}_{i,k},\, \boldsymbol{\delta}_{i,k}\big) \in \mathcal{A}_i(s_k), \tag{12}$$

where

$$\mathbf{p}_{i,k} = [p_{i,k}^1, \ldots, p_{i,k}^Q]^T, \quad \boldsymbol{\delta}_{i,k} = [\delta_{i,k}^1, \ldots, \delta_{i,k}^Q]^T. \tag{13}$$

Here, $p_{i,k}^q$ denotes the transmitted power allocated by radar node i to target q at time step k, and $\delta_{i,k}^q \in \{0,1\}$ indicates whether node i tracks target q.

The feasible action set $\mathcal{A}_i(s_k)$ is constrained such that each radar node can track at most one target and assign power only to the selected target:

$$\sum_{q=1}^{Q} \delta_{i,k}^q \le 1, \quad 0 \le p_{i,k}^q \le \delta_{i,k}^q\, p_{\max},\ \forall q. \tag{14}$$

The network-wide power budget imposes the constraint

$$\sum_{i=1}^{N}\sum_{q=1}^{Q} p_{i,k}^q \le p_{\text{total}}, \quad \forall k. \tag{15}$$

The instantaneous reward is designed to promote network-wide tracking accuracy. Specifically, a team reward shared by all players at time step k is defined as the negative sum of the PCRLBs over all targets,

$$r_k \triangleq -\sum_{q=1}^{Q} \mathrm{tr}(\mathbf{C}_k^q) = -\sum_{q=1}^{Q} \mathrm{tr}\big(\mathbf{J}^{-1}(\mathbf{X}_k^q)\big)\,, \tag{16}$$

The state-transition probability $P(s_{k+1} \mid s_k, a_k)$ is induced by the target motion model in (1); under the adopted formulation, it depends on the target dynamics and is independent of the radar actions.

B. Multi-Agent Reinforcement Learning

In MARL, the agents correspond to the players of the SMG. In the considered scenario, the objective is to optimize global tracking performance, which motivates the adoption of the CTDE paradigm.

Under CTDE, each radar node is equipped with a decentralized actor policy, while a centralized critic is used during training to incorporate global information. Proximal policy optimization (PPO) is adopted as the underlying policy gradient method. Let $\pi_{\theta_i}(a_{i,k}|o_{i,k})$ denote the policy (actor) of agent i. The PPO objective is given by

$$J_{\text{PPO}}(\theta_i) = \mathbb{E}\Big[\min\Big(\rho_{i,k}(\theta_i)\hat{A}_{i,k}, \tilde{\rho}_{i,k}(\theta_i)\hat{A}_{i,k}\Big)\Big]\,, \tag{17}$$

where

$$\rho_{i,k}(\theta_i) = \frac{\pi_{\theta_i}(a_{i,k}|o_{i,k})}{\pi_{\theta_i^{\text{old}}}(a_{i,k}|o_{i,k})}, \tag{18}$$

and $\tilde{\rho}_{i,k} = \mathrm{clip}(\rho_{i,k}, 1-\epsilon, 1+\epsilon)$. Here, $\hat{A}_{i,k}$ denotes the advantage estimate and ϵ is the clipping parameter.

The centralized critic $V_\psi(s_k)$ is trained by minimizing

$$L_V(\psi) = \mathbb{E}\Big[(V_\psi(s_k) - \hat{R}_k)^2\Big]\,, \tag{19}$$

where $\hat{R}_k$ denotes the empirical discounted return.

MAPPO extends PPO to multi-agent settings by employing decentralized actors conditioned on local observations and a centralized critic conditioned on the global state during training [20]. Hence, each actor uses only node-level information available after local preprocessing, without any inter-node information exchange. The centralized critic is conditioned on the global state s_k during training.

During training, the centralized critic observes the global state s_k and evaluates the joint action $\mathbf{a}_k = \{a_{i,k}\}_{i=1}^N$ to estimate the value function and compute the advantage $\hat{A}_{i,k}$ for each agent, where $\mathbf{a}_k$ denotes the joint action of all agents. Each decentralized actor updates its policy based on locally collected trajectories. The training process is conducted offline using simulated interactions with the environment. During execution, each radar node independently selects its action $a_{i,k}$ according to its learned policy $\pi_{\theta_i}(a_{i,k} \mid \mathbf{o}_{i,k})$ using only its local observation vector $\mathbf{o}_{i,k}$, without requiring any information exchange with other nodes.

IV. Numerical Results

A multistatic radar network with $N = 4$ radar nodes is considered for tracking $Q = 2$ targets under an IRF-equivalent clutter condition. Target 1 is initialized at position $(30, 30)\,\mathrm{km}$ with velocity $(-50, 300)\,\mathrm{m/s}$ and acceleration $(-1, 0.1)\,\mathrm{m/s^2}$, whereas Target 2 is initialized at position $(20, 10)\,\mathrm{km}$ with velocity $(100, 250)\,\mathrm{m/s}$ and acceleration $(0.5, 0.5)\,\mathrm{m/s^2}$. The complete scenario geometry is illustrated in Fig. 2.

For simplicity, the radar cross section (RCS) of all targets is set to $|\xi|^2 = 1\,\mathrm{m}^2$. The signal bandwidth is $\beta = 50\,\mathrm{MHz}$, and the single-beam illumination power is $1\,\mathrm{kW}$. The initial distance between each target and the reference base station is $50\,\mathrm{km}$, and the reference signal-to-noise ratio (SNR) is $10\,\mathrm{dB}$. The corresponding measurement-noise standard deviations are $\sigma_r = 100\,\mathrm{m}$ for range and $\sigma_\theta = 0.005\,\mathrm{rad}$ for bearing, and the process-noise intensity is set to $\kappa = 1\,\mathrm{m}^2/\mathrm{s}^3$.

For MAPPO, the policy is optimized with Adam using an initial learning rate of 10^{-4}, with a decay factor of 0.99. The PPO-related parameters are set to $\gamma = 0.95$, $\lambda_{\text{GAE}} = 0.95$, clipping coefficient 0.1, entropy coefficient 0.1, and value-loss coefficient 0.2, with gradient-norm clipping set to 2.0. For PSO, a threshold-based policy parameterization is adopted with a swarm size of 100, inertia weight $w = 0.7$, and cognitive/social coefficients $c_1 = c_2 = 1.5$.

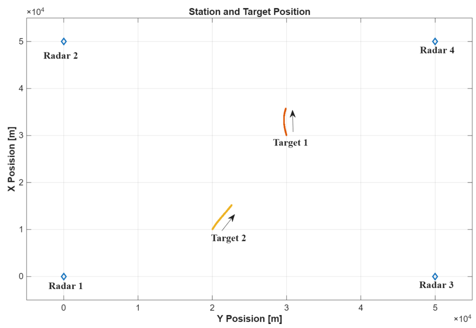

Fig. 2. The radar station distribution and target trajectories.

The following constraints are imposed in the considered scenario:

- Each radar station can only track a single target.
- For each tracking beam, the maximum transmitted power is set to $5\,\mathrm{kW}$.
- The total transmitted power across all radar nodes is constrained to $10\,\mathrm{kW}$.

To assess the performance of the proposed MAPPO-based sensing and power allocation scheme, two benchmark strategies and a particle swarm optimization (PSO)-based method are considered for comparison:

- *Benchmark 1*: Stations 1 and 2 track Target 1, stations 3 and 4 track Target 2, with $2.5\,\mathrm{kW}$ to each station.
- *Benchmark 2*: Stations 1 and 3 track Target 1, stations 2 and 4 track Target 2, with $2.5\,\mathrm{kW}$ to each station.

Performance is evaluated using both the sum of the PCRLBs and the position root mean square error (RMSE) of target tracking. These metrics jointly characterize the theoretical sensing performance and the practical tracking accuracy. Their time evolution is shown in Fig. 3 and Fig. 4.

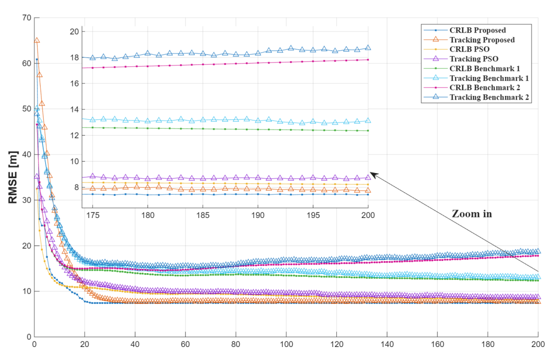

Fig. 3. RMSE, PCRLB and Tracking Result of Target 1.

Figures 3 and 4 depict the time evolution of the PCRLB and tracking RMSE for both targets under the proposed MAPPO-based method, the two benchmark schemes, and the PSO-based optimization approach. All results are averaged over 500 Monte Carlo simulations.

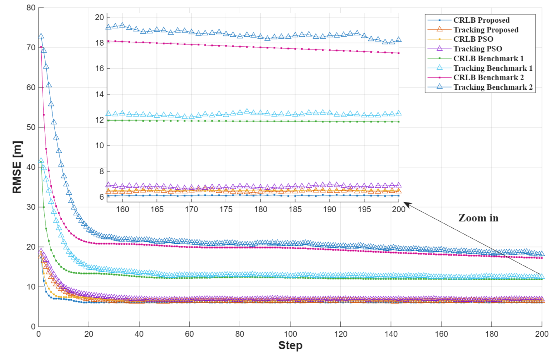

Fig. 4. RMSE, PCRLB and Tracking Result of Target 2.

For both targets, the proposed MAPPO method exhibits rapid convergence during the initial sensing stage, with both PCRLB and RMSE decreasing significantly within the first 20 time steps. This behavior indicates that the learned policy can quickly establish effective tracking geometries and allocate radar resources to the most informative stations. After convergence, MAPPO consistently achieves a lower steady-state RMSE than Benchmark 1, Benchmark 2, and PSO, demonstrating superior long-term tracking accuracy.

Although PSO attains comparable tracking performance, MAPPO offers a key advantage in terms of fully distributed execution. During deployment, each radar station independently determines target selection and power allocation based solely on local observations, without requiring global state exchange or centralized coordination. This enables parallel computation across stations, reduces communication latency, and enhances system robustness, making the proposed approach well suited for real-time and connectivity-limited scenarios.

V. Conclusions

In this paper, we considered a multistatic radar tracking network with the objective of enhancing tracking performance while reducing resource consumption, as quantified by the posterior Cramér-Rao lower bound. The problem was modeled as a stochastic Markov game, in which each radar station acts as an individual player that jointly performs node selection and transmitted power allocation to balance resource usage and tracking accuracy.

A centralized-training and decentralized-execution strategy based on multi-agent proximal policy optimization was adopted to solve the resulting optimization problem. Under the proposed CTDE-MAPPO framework, each radar station can execute actions using only its local observations, without requiring information exchange with other stations. This property supports low-overhead distributed execution while alleviating communication burden.

Numerical simulation results demonstrate that the proposed approach outperforms fixed assignment strategies and the PSO-based method in tracking accuracy, while its decentralized execution avoids iterative centralized optimization at each sensing step.

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